Data and tasks

4 downstream task definitions · patient-level 70/15/15 split

fixed

MIMIC-IV
74,829 processed stays

eICU-CRD
132,686 processed stays

Combined
207,515 processed stays

Each stay: $\mathbf{x}_i \in \mathbb{R}^{24\times 84}, \mathbf{m}_i \in \{0,1\}^{24\times 84}$

Common encoder architecture

fixed

Input
$[\mathbf{x}_{i,t}; \mathbf{m}_{i,t}] \in \mathbb{R}^{168}$

Tokenize
$\mathbb{R}^{168} \to \text{MLP} \to \mathbb{R}^{64}$

Transformer
2× pre-LN multi-head att.

Output
... $\mathbf{H}_i \in \mathbb{R}^{24\times 64}$

SSL implementations

varies

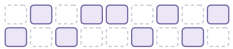
MAE
reconstruct masked hours

loss on input values

JEPA
predict teacher embeddings

loss on target embeddings

Contrastive
match projected views



loss on projected views

Baselines + extension

- **Supervised Transformer**
  same encoder, random init
- **GRU-D**
  recurrent, decay imputation
- **XGBoost**
  tabular summary features
- **TS2Vec-style extension**
  temporal-contrastive ext.

Evaluation

fixed

Downstream protocols:

Linear probe
task head only

Full fine-tuning
encoder + head

Tasks:

24-hour mortality
AUPRC

Hospital mortality
AUPRC

AKI KDIGO
AUPRC

Remaining ICU stay
Abs. error